Romy Mondello

AI AND FULL-STACK ENGINEER - AGENT INFRASTRUCTURE - DEVELOPER TOOLS

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AI, full-stack, and systems engineer with six years of professional software-engineering experience since joining Knox Financial in 2020. I build agent systems and the infrastructure they run on, with public work spanning JEV workflows for Codex and Claude Code, QQ runtime contributions, Rust systems, and formal verification. Earlier independent AI work began with deep learning in 2017 and language models around 2019, including GPT-3 beta participation.

Independent product engineering and open source

Independent Product Engineering / Open Source May 2025 - present

- Built JEV Workflows, a public open-source plugin for Codex and Claude Code that provides typed decision consultation, failure diagnosis, and evidence-based completion review. Model recommendations remain advisory and separate from permission to act.
- Contributed to the open-source Rust QQ coding-agent harness in merged PRs #333, #319, and #72, covering session-history recovery after held-tool cancellation or restart, shared JEV protocol validation, review checkpoints, and provider-neutral reasoning-effort settings.
- Directed coding agents through a QQ performance fix to conversation-history assembly, owning reproduction, testing, and independent review, while my cofounder added an indexed lookup. In a 300,000-event local benchmark, history assembly for 21 affected turns dropped from 1.98 s to 2 ms.
- Built Research OS, where agents and Z3 search and classify arXiv papers and route results through Aristotle for formal verification to certificates, running 60 to 90 agent-hours over two weeks.
- Co-developing a bounded local agent-execution prototype with a cofounder. It explores Temporal coordination, PostgreSQL run state, isolated workers, explicit cancellation controls, persisted evidence, and guarded verification.
- Built Ferrite, a public Rust-first TypeScript application-framework prototype with server-side rendering, file-system routing, artifact serving, WASM and Node bindings, and browser hydration.

Formal research Independent

- Authored a 10-page arXiv preprint giving a dimension-two counterexample to the separable Jacobian conjecture in characteristic two, with Jacobian determinant 1 and separable degree 3. Formalized the full theorem in Lean 4 and Mathlib with digest-locked sources and replay receipts.

Professional experience

Further - Consultant, Research Engineer (contract) Feb 10 - Apr 14, 2025

- Co-designed the LangGraph agent and data-pipeline architecture with the CTO, then built Zillow and mortgage API pipelines with typed state, tool orchestration, retrieval, and rule-based risk checks.
- Deployed the agent infrastructure on Kubernetes on Google Cloud with Nest.js and Python services and Azure as the inference provider, and stayed on as an advisor after the contract.
- Helped the team use Cursor and coding agents, and fixed LaTeX rendering and API timeouts in the AI Genius web app.

Pillzumi - Machine Learning Engineer (contract) Dec 2024 - Feb 2025

- Built long-running elizaOS social-agent integrations across WebRTC and X direct-message workflows, with persistent memory, action frameworks, and API-rate-limit handling. Agents posted 30 to 50 times a day and engaged roughly 50 to 100 people a day across DMs and public tags.

Verizon Business - Customer Solutions Apr - Dec 2024

- Owned a CRM book of business in a technical-consultative B2B motion. Pitched about 50 prospects a day, closed 4 to 6 accounts a week as a top closer, and trained other sellers.

Verus AI - Data Scientist Jan 2022 - 2023

- Built Google Cloud pipelines into BigQuery and Vertex AI, trained recurrent models for spend, usage, and customer-growth forecasting, and monitored model and data quality.

Flask - Founder & Lead Engineer Apr 2021 - Jan 2022

- Founded Flask, a fully decentralized DAO on the Avalanche blockchain, raised about $50K, and wrote the smart contracts myself.
- Hired and managed the whole team (a designer, a project manager, a small team of developers, and a Discord community manager) and gave the board of advisors regular updates.
- Grew a Discord community of roughly 500 to 1,000 members with heavy daily chat. An early adopter of AI-generated code in my workflow: it needed heavy editing then, but it still made me faster.

Knox Financial - Backend Software Engineer Jul 2020 - Mar 2021

- Built real-estate search on data feeds and AWS and Google APIs, distributed ingestion, PostgreSQL forecasting services, REST APIs, funnel instrumentation, and forecast visuals.
- Hardened CI/CD, reduced flaky-test noise, introduced deploy gates, and reviewed backend code.

Education

Associate of Science, Computer Science
Bunker Hill Community College - Dean's List

Technical

TypeScript, JavaScript, Python, Rust, Lean 4, Swift, Go | React, Next.js, Node.js, FastAPI, Axum | PostgreSQL, Temporal, Redis, Docker, AWS, GCP, BigQuery | LangGraph, OpenClaw, elizaOS, RAG, MLX, PyTorch/MPS